Physics-Inspired Quantum Error Mitigation by Generative Autoregression

A Novel Framework for Decoupling Coherent and Stochastic Noise

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Outline

Problem statement

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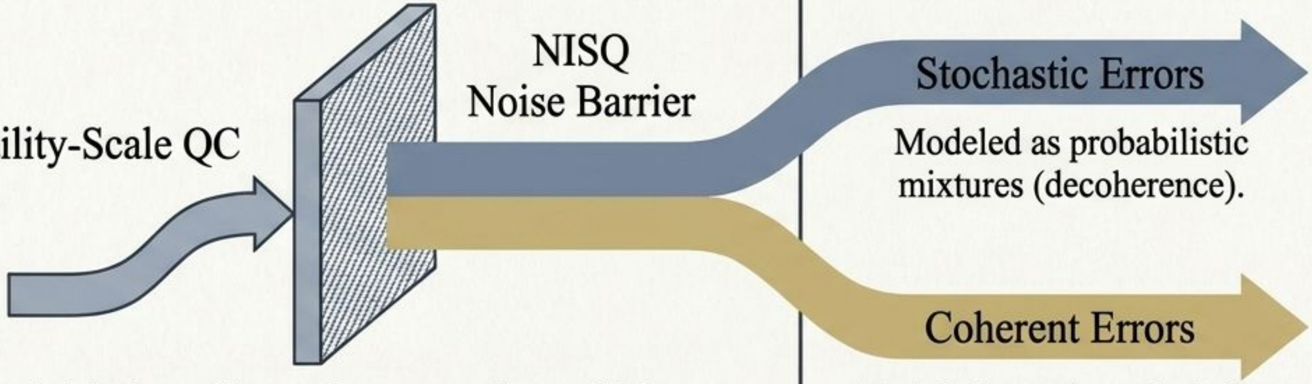
Introduction & Motivation

The Bottleneck



Achieving utility-scale computation on Noisy Intermediate-Scale Quantum (NISQ) devices requires advanced Quantum Error Mitigation (QEM).

The Physical Reality of Noise

Stochastic Errors

$$\mathbf{E}_l = (1 - p_l)\mathbf{I} + p_l \mathbf{\Lambda}_l$$

Modeled as probabilistic mixtures (decoherence).

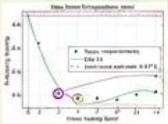
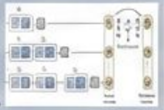
Coherent Errors

$$U_l^{\text{noisy}} = e^{i(H_l + \delta h_l)t}$$

Modeled as unitary deviations due to control imperfections or calibration drift.

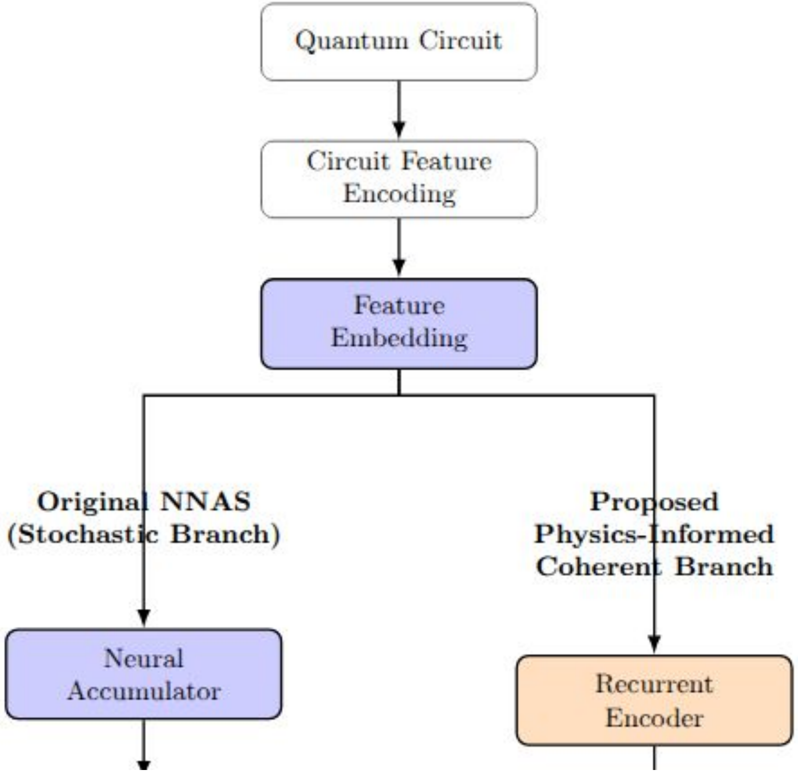
Key Insight: Stochastic and Coherent errors compound through entirely different physical mechanisms. Most current QEM models conflate them into a single monolithic noise representation.

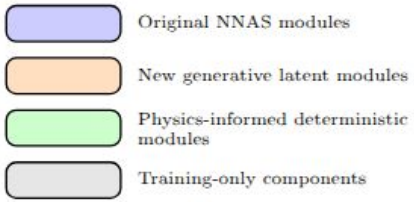
Related Work & The Theoretical Gap

Model	Data Efficiency	Generalizes to Circuit Depth	Explicit Coherent Error Tracking
Zero-Noise Extrapolation (ZNE) Simple, but biased and unstable with similar noisy samples. 	Linear Overhead	✗	✗
Unstructured ML-QEM Treats mapping as a black-box regression.	Poor	✗	✗
Original NNAS [9] Uses Maximum Noise Decomposition (MND) for layer-by-layer stochastic tracking. 	Excellent	✓	✗
Proposed Framework Maintains NNAS efficiency while actively decoupling coherent error accumulation.	Excellent	✓	✓

Architecture

- Quantum circuit describe computation
- An MLP encodes device and circuit configuration
- Neural Accumulator (NNAS - Incoherent errors)
  Single hidden state RNN to model noise accumulation
- Recurrent Encoder (new - Coherent errors)
  Gated Recurrent Unit (GRU) to model noise accumulation

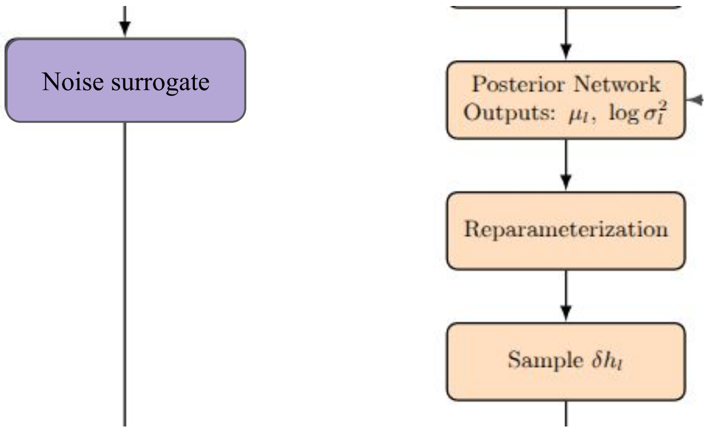


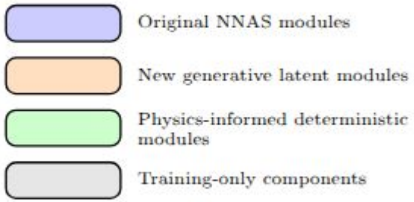


Architecture

- Surrogate object (NNAS - Incoherent errors)
    Predicts noise accumulation for every layer in the quantum circuit.

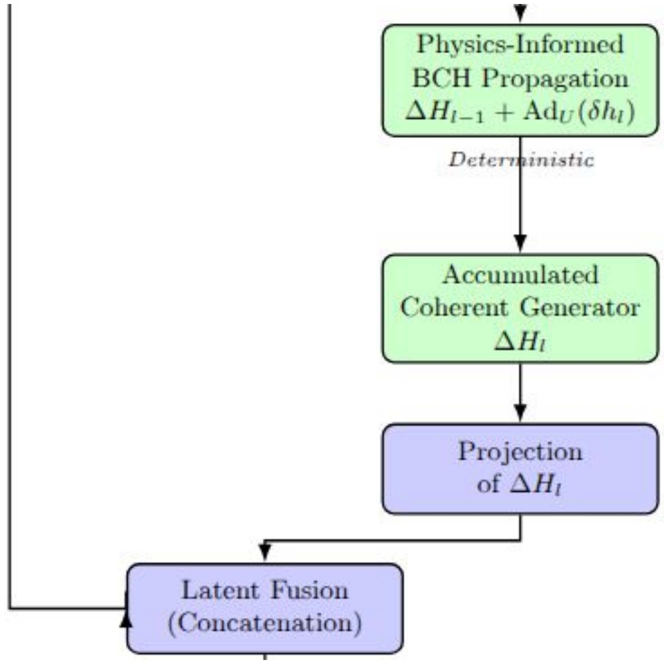
- Latent space (new - Coherent errors)
    - Models a Gaussian latent space the noise generator δh_l
    - δh_l governs the physical dynamics of coherent noise
    - Expecting the latent to better generalize than regression

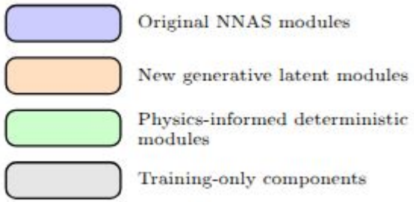




Architecture

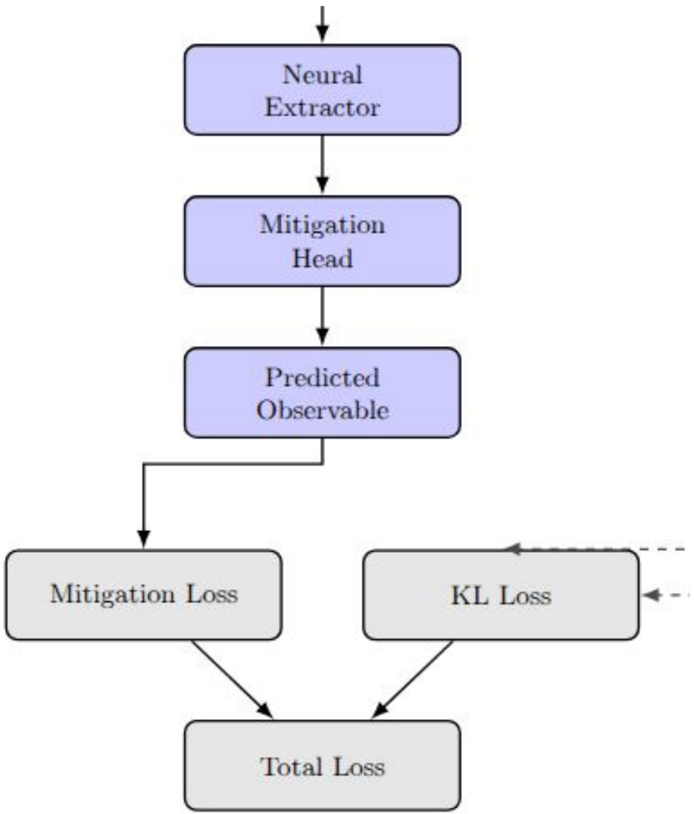
- Recurrent Encoder (new - Coherent errors)
  - Coherent can be exactly evolved given δh_l
  - Accumulate noise by the BCH approximation (no parameters)
  - Learned projection and concatenate

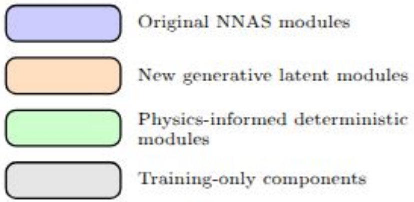




- Same attention based neural extractor as with NNAS
  - The symmetries of attention resemble quantum noise accumulation
  - Neural extractor: decoder

- Additional KL loss term to train the latent.



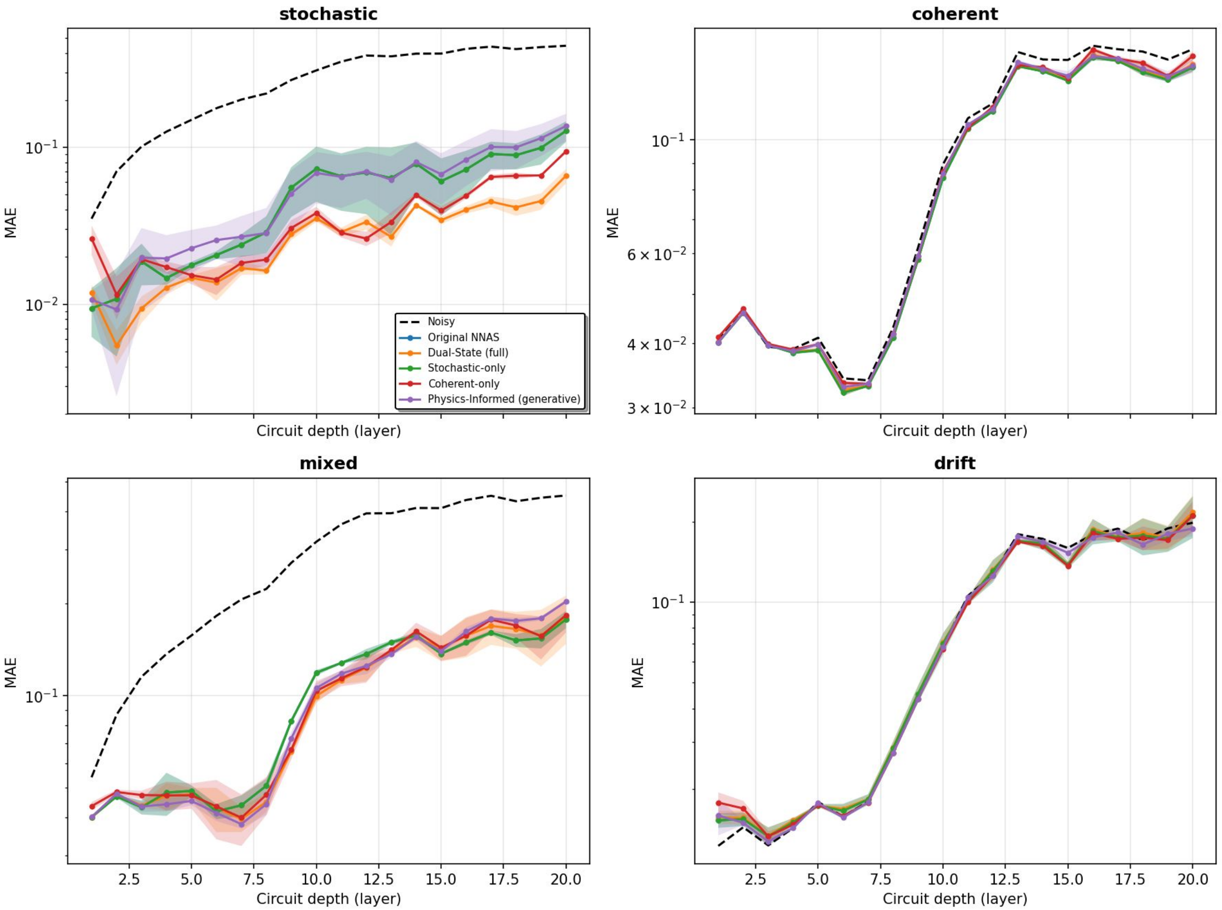


Experimental results

Proof-of-concept experiments
No supremacy claims

5 model variants
4 different datasets

Takeaway: adding more physical guidance does not degrade the mitigation performance



No pre-existing code, model or datasets :)

Limitations & Future Directions

Current Proof-of-Concept Constraints

Currently uses basic implementations for hyperparameter optimization and prior distributions.

Algorithmic Next Steps

Refine priors with precise data and improve the training pipeline.

Scaling to Quantum Utility

Evaluate on larger datasets and deploy on real-world quantum hardware.

Key Contribution

Establishing a new principled foundation for combining generative machine learning with physically constrained models of quantum noise.